

Multiple Choice

1. **Answer:** D

Options: A) magnetic flux through S; B) rate of change of the magnetic flux through S; C) time integral of the magnetic flux through S; D) rate of change of the electric flux through S

Note: Correct option: D - rate of change of the electric flux through S

2. **Answer:** A

Options: A) 4; B) 5; C) 6; D) 20

Note: Correct option: A - 4

3. **Answer:** C

Options: A) There are no changes in the internal energy of the system.; B) The temperature of the system remains constant during the process.; C) The entropy of the system and its environment remains unchanged.; D) The entropy of the system and its environment must increase.

Note: Correct option: C - The entropy of the system and its environment remains unchanged.

4. **Answer:** B

Options: A) $n = 3, l = 2$; B) $n = 3, l = 1$; C) $n = 3, l = 0$; D) $n = 2, l = 0$

Note: Correct option: B - $n = 3, l = 1$

5. **Answer:** D

Options: A) 1; B) 2; C) 3; D) 5

Note: Correct option: D - 5

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '250'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hall coefficient'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $2 V_0/3$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.60 c. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key